Fourth addendum to the 3.0.9/3.1.1 Release Notes

November 6th, 2024

A series of code improvements, including some important bug fixes and adaptations required by the migration of the SDCC cluster to Red Hat 9 architecture and usage of IMAS Access Layer 5, have been made to the SOLPS-ITER 3.0.9 *master* branch following the Prague Code Camp, which triggered this update to the SOLPS-ITER master version. Unless mentioned otherwise, the same changes have also been applied to the 3.0.9 *develop*, *release/3.1.1* and *feature/wg_workflow* reference branches. As requested during the Code Camp, I have also done some extensive pruning of obsolete branches in the SOLPS-ITER repository and associated submodules.

The changes of note include:

1. When compiling SOLPS as a whole, the main *Makefile* will first compile the netCDF utility programs **nc2text_simple** and **nc_reduce**, using **B2.5** stand-alone options with no parallelization, and then proceed with the other targets being asked for. If the user invokes, before launching the compilation, any of the *(un)set_mpi*, *(un)set_openmp*, or *(un)set_debug* environment setup scripts, then only the executables with those options will be compiled. This means that `set_mpi ; gmake solps` is now equivalent to `gmake solps_mpi`. The **Uinp** compilation is now set to need the corresponding `b25eirene` target as a prerequisite.
2. The 3.0.9 *master* code is now linked to **Eirene** MsV version 1.0.10. This notably includes a new coding rules document and a *CHANGELOG* file.
3. A superfluous 10^6 conversion factor had crept in during the last master update when writing out the atom temperature array `tab2` from **Eirene** to the *fort.44* file. This is now fixed. This bug did not affect the Wide Grids code version.
4. Several environment files were updated according to the user reports provided to me during the Code Camp, and compilation flag options adapted to the newest GNU (13.2.0) and Intel (2021 & 2023) compilers, always aiming to have as warning-less a compilation as reasonably achievable. The ITER environment files have been modified to suit the new Red Hat 9 architecture on the SDCC cluster, but with backward compatibility to the older remaining CentOS8 nodes. Several of the CI test cases were also rebuilt. Some associated minor code clean-up also took place, as well as modularization of a large number of common blocks throughout the code suite. The **b2sxdr** compilation flags were also modified to allow for better handling of the different precision requirements between the SOLPS-ITER and SOLPS4.3 components.
5. Hecko Jan EXT has provided `CONTAINER` configuration files for use inside SOLPS-ITER images. He also introduced two new environment variables, `SOLPS_COMPILER_FORCE` and `SOLPS_HOST_NAME_FORCE`, which can be used to override the settings that the *default_compiler* and *whereami* scripts would provide (or replace them if they would fail, like inside a container). The installation instructions have been updated to mention these variables and show their use.
6. The code was adapted to the IMAS Data Dictionary 3.42.0 standard and now uses IMAS Access Layer 5.3.0. The switch `b2mndr_shot_number` is renamed `b2mndr_pulse_number` to follow Access Layer 5 nomenclature. It is also now possible to specify a path instead of shot and run numbers to indicate where the IMAS data entry is to be read or written, using the `b2mndr_ids_path` variable. The `--shot` argument for the ids2dg and dg2ids conversion programs is now renamed `--pulse`.
7. A bug fix provided by Hecko Jan EXT is implemented to get the gas puff feedback boundary condition `BCCON=11` to work when the gas injected is a neutral species. This bug did not affect the Wide Grids code version.
8. Leveraging the work of Hecko Jan EXT , the code (3.0.9 and 3.1.1 versions) can now use indifferently the fixed or JSON formatted **Eirene** input files. The various coupled programs now look for the *fort.1* file, which is a symbolic link to the chosen format of the input file, according to whether the run directory was set up using the *setup_baserun_eirene_links* or the new *setup_baserun_eirene_json_links* script, respectively. Similarly, the *runs/Makefile*, job submission and file management scripts have been adapted accordingly.
9. If the **Eirene** input file does not contain a time-dependent stratum (i.e. `NPRNLI` in block 13 is set to zero), then, if a `'T'` stratum is defined in *b2.neutrals.parameters*, the contradiction is resolved by ignoring that latter `T` stratum and thus ensuring that both **B2.5** and **Eirene** are handling the same number of strata. Conversely, if the `T` stratum is not defined in *b2.neutrals.parameters* but `NPRNLI > 0` in block 13 requests it, then it is automatically added.
10. A bug fix, provided by Lore Jeremy EXT , avoids the untimely closing of the *run.log* file by **Eirene** when run in parallel, which could happen for certain compiler implementations. This bug affected the **Eirene** MsV implementation.
11. In the **Eirene** submodule, we now enforce the naming convention that source files with a *.F* extension are meant to be pre-processed before compilation, while *.f* extension files do not need such a step. In parallel, a series of pre-processor pragmas have been added in the **Eirene** code source as well, to reduce the number of compiler warning messages from the latest compiler updates.
12. The code now enforces that **Eirene** MsV cannot be run with both the `USE_OPENMP` and `USE_EXT_OPENMP` options simultaneously, which represent multi-threading of the code inside or outside the **Eirene** call respectively, and are mutually exclusive.
13. A bug fix was implemented for **Eirene** MsV OpenMP runs when multiple threads are accessing the `SAMVOL` routine with differing lists of recombination reactions.
14. A missing definition for the optional **Eirene** emissivity tally was corrected.
15. To increase the stability of the solver when running with OpenMP, and improve result reproducibility for CI testing, the solver matrices are now rebuilt on every call and no longer rely on a previous iteration having done so.
16. When building a grid with **Carre**, it now checks that the total number of radial and poloidal cells that the grid will end up having are within the limits provided by the *b2mod_dimensions.F* file (respectively *carre_dimensions.F* when running in stand-alone mode). If the provided array dimensions are too small, the user is told to reduce the grid resolution or increase the declared array sizes and recompile.
17. Following a bug report by Jung Dami EXT , a minimum value of `1` is now enforced for the `nncutmax` variable, to ensure the **Carre2** and **b2ag** algorithms work properly.
18. The default values of the `b2ux9p_mult_nonzero` and `b2ux11p_mult_nonzero` were increased from `10` to `15` to make sure a sufficient workspace is provided to the solver for the 9-point-stencil mode.
19. A bug was fixed in the write-up of the batch *edge_profiles* IDS, to use the time-averaged parallel velocity instead of the snapshot one.
20. A sturdier test, provided by Carli Stefano EXT , checks that the SOLPS session is started inside a `tcsh` shell.
21. Some bug fixes to the *last_2d* and *plot_trc* scripts were provided by Pshenov Andrei .
22. A `clean_manual` target now exists in the **Eirene** SOLPS-style *config/Makefile* to help when rebuilding the **Eirene** manual from scratch. Problematic characters that were preventing a clean SOLPS Manual build were removed from the underlying source material and/or are now filtered out by the *Makefile* creating it.
23. The inline documentation and main Manual were updated to reflect the changes above, as well as include wording clarification and correction suggestions from various users.

Extra notes for installers/developers:

- Point 1 above also means that the parallelization options are now separated in the *Makefile*s from the serial compilation options. Parallel-specific options are now stored in the `FPOPTS` and `LPOPTS` variables, for the compiler and linker respectively, while the other (serial) options remain in `FC`

`OPTS` and `LDOPTS` respectively. The preprocessor definitions are in `DEFINES` for serial compilations and the parallel-specific ones are stored in `DPFINES`. Both sets are used together for parallel compilations, so no need for duplication. The serial compiler and linker to use for the netCDF programs compilation are stored in the `FN` and `LN` variables respectively. In cases where the MPI compiler is a GNU compiler but the serial compiler is an Intel one, the *Makefile* variable `COMPILER_REDEF` is used to switch to the correct set of flags. Also make sure the `MPI_FC` variable (indicating what your MPI compiler is) is declared in your *$SOLPSTOP/SETUP/setup.csh.$HOST_NAME.$COMPILER* file. If not defined in that same file, the `FC`, `FN`, `LD`, and `LN` variables will be set by the *$SOLPSTOP/SETUP/config.common.$COMPILER* files.

- The `SOLPS_OPENMP` and `SOLPS_MPI` environment variables are meant for usage inside the *Makefile*s, while the pre-processor pragmas `USE_OPENMP` and `USE_MPI` are meant for the source code.
- I have corrected the **Eirene** *CMakeLists.txt* file to be able to accept additional local dependencies, to be put in *src.local/CMakeLists.txt*, with the same syntax as the example below:

  ```
  set_source_files_properties(${PROJECT_SOURCE_DIR}/output/find_emis_line.f PROPERTIES OBJECT_DEPENDS
  ${LOCAL_OBJDIR}/output/emissivity.f.o)
  ```
  where the first source file is the calling routine and the second one the object file of the called routine.
- The CI test scripts now do a check with both the **Eirene** fixed and JSON input formats, as well as testing the `gfortran` toolchain.
- The reassignment of common blocks into modules is as follows:
  - In **B2.5**:
    - *src/common/COUPLE/EIRSRT.F src/modules/b2mod_eirsrt.F*
    - *src/common/eir_reac.F src/modules/b2mod_eirreac.F*
    - *src/common/graphics_vars.h src/modules/b2mod_graphics.F*
    - *src/common/rdload.inc src/modules/b2mod_rdload.F*
    - *src/common/stackw.h src/modules/b2mod_stack.F*
    - common blocks from inside *src/utility/adpak.F src/modules/b2mod_adpak.F*
    - `b2ngeom` common block absorbed into *src/modules/b2mod_neoclassical.F*
    - `chord_shift` common block *src/modules/b2mod_chord_shift.F*
    - `line_format` and `t_ratio` common blocks absorbed into *src/modules/b2mod_b2plot.F*
    - `zoom_com` common block absorbed into *src/modules/b2mod_graphics.F*
  - In **Carre**:
    - *src/include/CARREDIM.F src/include/carre_dimensions.F*
    - *src/include/CARRETYPES.F src/include/carre_types.F*
    - *src/include/COMLAN.F src/include/comlan.F*
    - *src/include/COMORT.F src/include/comort.F*
    - *src/include/COMQUA.F src/graphe/comqua.F*
    - *src/include/COMRLX.F src/include/comrlx.F*
    - *src/include/FCRCOM.F src/include/fcrcom.F*
    - *src/include/PERIM.F src/graphe/perim.F*
    - `geom` common block absorbed into `comort`

To obtain the new 3.0.9 *master* version, use the *solps-iter_update* script. Be mindful that, because many of the setup and configuration files change with this update, the code will most likely fail to compile the first time around. In that case, close your session, restart it to load the updated setup and configuration files, and attempt the compilation again. Same story for the 3.0.9 *develop*, *release/3.1.1* and *feature/wg_workflow* branches, but using the *solps-iter_update_develop*, *solps-iter_update_3.1.1_release*, or *solps-iter_update_wg_workflow* script, respectively, of course.